

WYNYARD, Australia

WMO: 959570

Lat: 40.997S

Lon: 145.729E

Elev: 39

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.7	33.4	29.8	23.9	40.8	31.3	25.7	40.5	27.6	52.2	25.2	51.5	6.7	280	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.6	75.7	63.0	72.9	62.8	70.7	61.6	67.0	71.5	65.5	69.7	64.1	68.4	13.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	93.7	68.8	63.7	88.5	68.1	62.2	83.9	67.0	31.5	71.6	30.5	69.8	29.4	68.5	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.5	20.3	DB	28.6	84.5	1.1	4.5	27.8	87.7	27.2	90.3	26.6	92.8	25.8	96.0	(4)
(5)			WB	28.8	69.5	1.5	2.6	27.8	71.4	26.9	72.9	26.1	74.4	25.0	76.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	61.2	61.7	58.6	54.1	50.4	47.7	46.6	47.1	48.9	51.5	55.3	58.0		
	DBStd	6.77	4.69	4.56	4.83	4.57	4.11	4.14	3.85	3.78	3.71	3.96	4.48	4.39		
	HDD50	485	0	0	2	14	46	95	120	106	64	30	7	1		
	HDD65	4292	137	109	205	326	454	520	570	555	484	418	293	221		
	CDD50	1714	346	329	268	138	58	24	14	17	30	76	165	249		
	CDD65	46	18	17	6	0	0	0	0	0	0	0	1	4		
	CDH74	197	99	46	14	1	0	0	0	0	0	0	11	26		
	CDH80	31	22	5	1	0	0	0	0	0	0	0	1	2		
(14) Wind	WSAvg	10.1	11.1	10.6	10.0	9.1	9.2	8.6	9.4	9.4	10.4	10.5	11.2	11.3		
(15) Precipitation	PrecAvg	37.8	1.9	1.9	1.6	2.6	3.4	4.3	5.6	4.1	4.3	3.3	2.6	2.2		
	PrecMax	53.2	6.5	5.7	5.0	6.9	8.1	7.5	9.8	7.6	7.6	5.6	4.4	6.1		
	PrecMin	30.8	0.2	0.2	0.2	0.5	0.7	1.5	1.2	2.3	1.4	1.2	0.6	0.4		
	PrecStd	6.0	1.4	1.5	1.2	1.6	2.1	1.8	2.3	1.5	1.8	1.1	1.0	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.4	78.9	75.7	69.6	63.8	60.3	58.4	59.9	62.9	69.1	74.5	77.7		
		MCWB	64.0	64.3	65.5	60.2	57.4	55.5	53.0	52.0	54.0	57.2	60.5	61.4		
	2%	DB	75.9	74.5	72.0	66.6	61.6	58.4	56.6	57.7	60.1	64.6	69.0	72.0		
		MCWB	62.6	63.8	63.0	59.8	56.7	54.1	52.1	51.8	53.0	55.0	58.7	60.9		
	5%	DB	72.6	72.3	69.5	64.6	60.2	57.1	55.5	56.2	58.3	61.9	66.3	69.2		
		MCWB	62.5	62.9	62.2	58.7	55.8	53.3	51.5	51.3	52.3	54.0	57.8	59.3		
	10%	DB	70.1	70.2	67.7	63.1	58.9	55.9	54.4	54.8	56.6	59.9	64.0	66.8		
		MCWB	61.2	62.2	60.9	57.8	55.0	52.4	50.8	50.7	51.7	53.5	56.9	58.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.6	68.8	67.9	63.4	60.1	57.4	55.4	55.3	56.8	59.5	63.9	66.1		
		MCDB	76.4	72.7	72.4	66.2	61.3	59.0	56.4	56.8	59.1	64.0	68.3	71.6		
	2%	WB	66.6	67.0	65.8	61.9	58.5	55.5	53.5	53.5	55.2	57.9	61.6	63.8		
		MCDB	71.2	71.1	69.7	64.7	60.4	57.2	55.3	55.8	57.9	61.5	66.2	68.1		
	5%	WB	64.9	65.7	64.1	60.4	57.1	54.2	52.4	52.4	53.9	56.3	59.7	62.3		
		MCDB	69.5	69.8	67.8	63.8	59.4	56.4	54.7	55.1	56.8	60.0	64.2	66.4		
	10%	WB	63.2	64.3	62.4	58.7	55.5	52.9	51.4	51.4	52.6	54.8	58.1	60.6		
		MCDB	68.1	68.5	66.5	62.5	58.4	55.5	54.0	54.3	55.6	58.8	62.8	65.2		
(35) Mean Daily Temperature Range	MDBR		16.5	15.6	16.6	16.6	15.7	15.0	14.6	14.9	15.1	16.1	15.6	16.4		
	5% DB	MCDBR	20.8	17.9	17.6	17.5	16.0	14.5	14.3	15.5	16.9	18.7	19.3	20.4		
		MCWBR	11.2	10.1	10.9	11.6	11.7	10.7	10.4	11.0	11.0	11.4	11.4	11.5		
	5% WB	MCDBR	15.3	13.1	14.5	14.8	13.3	12.0	11.7	13.0	13.8	15.6	15.0	15.1		
		MCWBR	10.4	9.1	10.8	11.1	10.9	9.7	9.5	10.3	10.4	11.2	10.3	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.346	0.334	0.326	0.315	0.307	0.308	0.324	0.348	0.343	0.343	0.349		
	taud		2.498	2.538	2.573	2.571	2.580	2.587	2.588	2.545	2.462	2.492	2.515	2.505		
	Ebn at Noon		309	303	293	274	255	247	255	270	283	301	311	312		
	Edh at Noon		36	32	29	25	21	19	20	25	31	34	35	36		
(44) All-Sky Solar Radiation	RadAvg		2135	1853	1395	951	610	484	527	776	1131	1613	1947	2103		
	RadStd		137	91	67	61	46	45	31	62	79	96	127	145		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+0.83	N/A	N/A	+1.62	+0.92	+0.88	N/A	-245	+230
(47) Regional (18 neighbors)	DBAvg	+0.49	N/A	N/A	+1.17	+0.74	+0.81	N/A	-144	+162

Nomenclature: See separate page